

MINIPROJECT REPORT

ENVIRONMENTAL ISSUES OF WIND TURBINES: EFFECT ON MICROCLIMATE AND BIRDS



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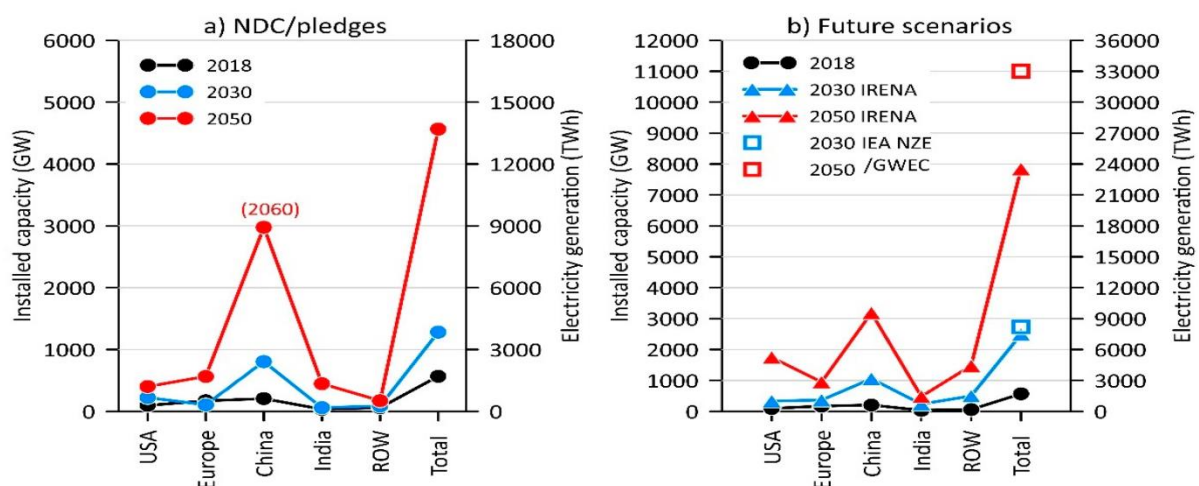
ENVIRONMENTAL ISSUES OF WIND TURBINES: EFFECT ON MICROCLIMATE AND BIRDS

ABSTRACT

Recognized as one of the most mature renewable energy technologies, wind energy has been developing rapidly in recent years. Many countries have shown interest in utilizing wind power, but they are concerned about the environmental impacts of the wind farms. The continuous growth of the wind energy industry in many parts of the world, especially in some developing countries and ecologically vulnerable regions, necessitates a comprehensive understanding of wind farm induced environmental impacts. Marine birds in particular have life history traits that amplify population impacts from displacement and collision with offshore wind infrastructure. Here, we present a broadly applicable framework to assess and mitigate the impacts of OWED (Offshore and Wind Energy Development) on marine birds. The environmental issues caused by wind farms were reviewed in this paper by summarizing existing studies. The potential effects of the proposed increase in wind energy developments on birds are explored using information from studies of existing wind farms. Evidence of the four main effects, collision, displacement due to disturbance, barrier effects and habitat loss, is presented and discussed. Bird mortality due to collisions with wind turbines is one of the major ecological concerns associated with wind farms. Data on the factors influencing collision risk and bird fatality are sparse and lack integration. This baseline information is critical to the development and implementation of effective mitigation measures and, therefore, is considered a priority research topic. The consequences of such effects may be direct mortality or more subtle changes to condition and breeding success. The requirements for assessing the impact of future developments are summarized and a particular emphasis on the rapidly developing area of offshore wind farm assessments. Wind turbines impact the local environment due to generation of downstream wakes (areas of disturbed flow behind each wind turbine). Like the water wake behind a motorboat, wind turbines create a wake of slower, more choppy air that eventually spreads and recovers its momentum. The wake increases mixing in the lower atmosphere and thus wind turbines have an impact on near-surface air temperature and humidity. While converting wind's kinetic energy into electricity, wind turbines modify surface-atmosphere exchanges and the transfer of energy, momentum, mass and moisture within the atmosphere. These changes, if spatially large enough, may have noticeable impacts on local to regional weather and climate. Mitigation measures which have the potential to minimize impacts are also summarized. Finally, recent developments in the monitoring and research of wind energy impacts on birds are outlined and some areas for future work are described. The intention of this paper is to provide state-of-the-art knowledge about environmental issues associated with wind energy development as well as strategies to mitigate environmental impacts to wind energy planners and developers. We conclude that wind energy has an important role to play in future energy generation, but more effort should be devoted to studying the overall environmental impacts of wind power, so that society can make informed decisions when weighing the advantages and disadvantages of particular wind power development.

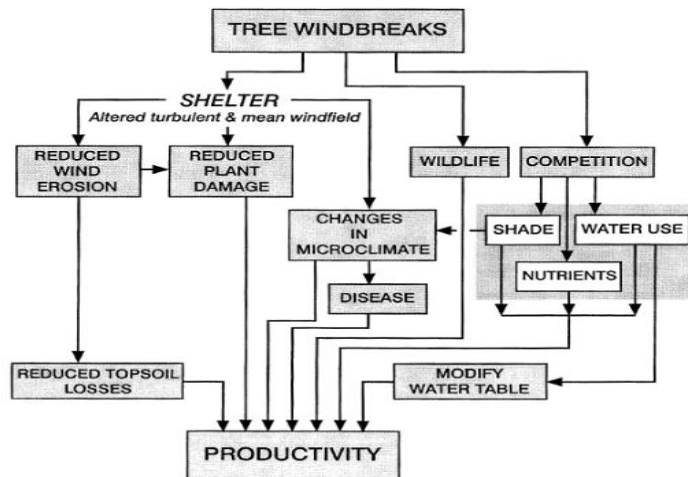
BIBLIOGRAPHIC RESEARCH

Wind energy is universally viewed as one of the most encouraging substitutes to fossil fuels because it can remarkably assist to depletion in greenhouse gas. The IPCC (Intergovernmental Panel on Climate Change) forecasts that by 2050 electricity supplied from wind farms can comprise of 20% more than the international electricity contribution. Effectively, the elevated turbulence happens because of the wind turbine performance impacts the blending methods which can minutely interrupt the temperature and pressure circulations downstream of wind farms and may critically affect the indigenous ecosystems such as the prominent mastic tree inhabitants. [1] The speedy widening of wind energy has uplifted solicitudes about the effect of wind turbines on birds and microclimate. Influence on regional climate are extremely critical in circumstances where assigning large number of turbines is accepted by minimising carbon outflow, because these consequences could have different impacts on the carbon cycle in the altered area. A huge, suitably sited wind farm may destroy fewer birds than a poorly sited and small one. Onshore conventional wind turbines have been documented to cause warming at the surface during stable atmospheric conditions. [2] In stable conditions, vertical motion is suppressed and the lapse rate is sub-adiabatic; in unstable conditions, vertical motion is enhanced and the lapse rate is super-adiabatic; and in neutral conditions, vertical motion is neither suppressed nor enhanced and the lapse rate is adiabatic.



Onshore, stable conditions mainly happens in winter or at night time but offshore they are more likely to happen in summer, regardless of day or night. At first, the accepted technique to elucidate the temperature changes was that the high turbulence in the wakes would reach the ground and cause an increase in vertical mixing, leading to warming and cooling near the surface in stable and unstable conditions, respectively. It is a difficult task to directly measure the effects of wind farms on the local environment, even more so offshore, because of the multi-scale character of the turbulence from the turbines, the dynamic nature of the wakes, and the relatively large horizontal and vertical extents. [2] On the other hand, resolving explicitly every day that may form around every individual wind turbine in a wind farm is an almost impossible task due to the excessively high computational demand of such simulations. Thus, to leverage the power of sophisticated simulations without excessive computational requirements, Parameterization approach must be used, which is a simplified treatment that considers the spatially-aggregated impact of wind turbines on the resolved variables. [3] Several studies showed that enhanced turbulence produced by wind farms has impact on heat and moisture exchange between the surface and the atmosphere. In stable

atmospheric conditions, when a warm air layer is present above cooler air, increased turbulence results in a warming near the surface by transferring thermal energy from higher levels to the lower. On the other hand, when unstable atmospheric conditions are present, i.e. cool air lying above a warmer layer, enhanced turbulence leads to mixing the layers and provides a cooling effect on the ground. These impacts depend on the ratio of natural turbulence to turbulence generated by the wind turbine. The stability of atmospheric layers can also influence a wind farm's impacts. [3] The effect of vertical mixing can be 10–15% stronger when a wind turbine is present in the atmospheric boundary layer. In case of impact on birds it was examined that small numbers of mortalities per year at a wind farm would be regarded important, except few of those mortalities were of endangered creatures, in which case the results might happen at the population level. As the space of the farm gets enlarged (density of turbines remain fixed and unchanged), the possibility for unfavourable outcomes, other than mortalities, also enlarges. [4] Large provisions destroy more habitats of birds. Birds lose their homes and they need to compromise, so that pasture and reproducing birds may be more liable to escape the area. Wind turbines can affect the birds directly or indirectly. The collisions between birds and wind turbines cause the mortality which can be termed as the direct impact while the indirect impact consists of birds ignorance and loss of habitat due to wind farm performance and construction. The mortality of animals was thought to be much more if the zone selected for the wind farm is abundant in wildlife or belongs to the migratory track of birds. [4] The mortalities due to direct collision of the birds with the rotor blades of a wind turbine, the avoidance or displacement of animals due to the construction process and the noise produced by the wind turbines, the persistence of disturbance, and habitat loss caused by the construction of wind farms and their associated infrastructure are all proved to have a negative impact on wildlife. Wind farms can affect local climates in two main ways. The first is by extracting kinetic energy, which slows wind flow. [5] The second is by further heightening turbulence flow downwind. Turbulence can be generated in the wake of rotors and also by the shear between the non-affected faster air layer below the turbine and the upper air layer slowed by the turbine. Increasing bird mortality from direct collision with the blades of functioning turbines has become a crucial issue questioning the apparent environmental friendliness of wind energy. While it is important to reduce climate change, a transition to renewable energy should be made responsibly with negligible impacts to wildlife. [3] High-resolution simulations are carried out with the Weather Research and Forecasting (WRF) model in order to quantify downstream effects on near-surface climate and the differences in climate effects that arise from two different parameterizations in WRF (Fitch and the newly developed EWP scheme). Other studies have found that turbine height was negatively correlated with bird count, but the present study separated height from blade length and found length to be the more important factor.



The study's findings can be used to inform decisions about wind turbine placement and design. Because the impact on birds diminishes as the distance increases, the researchers suggest that wind turbines be placed outside a 1,600 meter buffer zone of high-density bird habitats. They also recommend that turbines be taller but with shorter blade length. Bat fatalities have emerged as a substantial issue with repowering from old-generation wind turbines to modern wind turbines in the APWRA. We therefore sought to determine whether operational curtailment can reduce bat fatalities in the APWRA and whether this mitigation measure could also reduce bird collision fatalities. High fatality detection rates in 2 recent studies enabled us to test the effects of operational curtailment on reducing fatalities of volant wildlife. [5] Despite over a decade of research on bat fatalities at wind farms, relatively little is known about why wind turbines kill bats. Observations using infra-red imagery and satellite tagging have both determined that bats appear to interact with turbines, which may heighten collision risk. [6] A scientist have found that female *Nyctalus noctula* repeatedly came into close contact with wind turbines during foraging flights and flew at heights that suggested a high risk of colliding with turbine blades. Bats interact with turbines and changing their orientation when approaching them depending on wind speed. For tree-roosting bats, this may be a consequence of individuals mistaking turbines for potential roosting sites or bats attempting to establish mating sites at turbines. However, this is unlikely to be the primary explanation in Europe, where most casualties are not tree-roosting species. Increased prey availability around turbines is also a hypothesis for increased bat presence around turbines. Although evidence suggests bats are attracted to large turbines, experimental assessment at small wind turbines found that bats actively avoided them. [6]

OBJECTIVES:

The aim of understanding the effect of wind turbines on microclimate is to learn how these renewable energy structures causes a change to the local weather and environmental surroundings. Here are some important objectives:

1. Assessing Wake Effects:

- A. The wake effects created by wind turbines downstream need to be examined.
- B. How turbulence generated by turbine operation influences mixing processes need lots of attention.

- C. The propagation of these wake effects under different turbulence intensities needs investigation.
- D. Wind farms may or may not affect the microclimate in specific regions and it should be determined. [3]

2. Evaluating Impact on Natural Ecosystems:

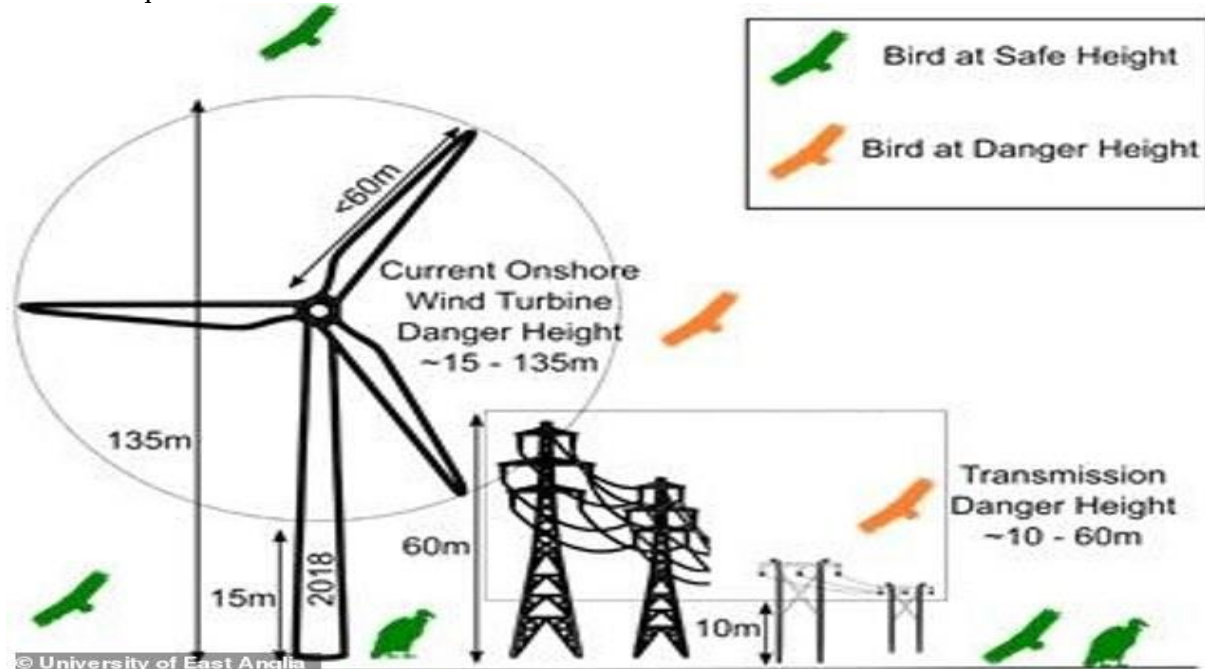
- A. The impact of wind farms on vegetation, tree populations and other forms of natural ecosystems needs a thorough examination.
- B. Wind turbines do alter local-scale meteorology and microclimates should be checked accordingly.

3. Quantifying Changes in Surface-Atmosphere Exchanges:

- A. The way wind energy alters surface-atmosphere exchanges of energy, momentum, mass, and trace gases should be examined.
- B. The unintended consequences of wind energy development on microclimate should be carefully observed. [3]

Hence forth investigating wind turbine effects on microclimate helps to involve sustainable energy practices and confirms minimal disruption to local ecosystems and weather patterns.

The aim of examining the impact of wind turbines on birds is to evaluate both the negative and positive effects of wind energy facilities on bird populations. Some of the basic addressed questions in some studies includes:



1. Do Wind Turbines Affect Bird Abundance?

- A. Whether wind turbines have an impact on bird population size and distribution have undergone a thorough research process by the researchers.

- B. The researchers estimated whether bird abundance alters in areas with wind farms in contrast with the reference areas. [4]

2. Vulnerability of Bird Taxa:

- A. Scientists tried to find out if certain bird taxa (such as passerines, raptors waterfowl, etc.) are more unsafeto wind turbine impacts.
- B. Exploringwhich species of birds most endangeredand inform conservation efforts.

3. Turbine Density and Power:

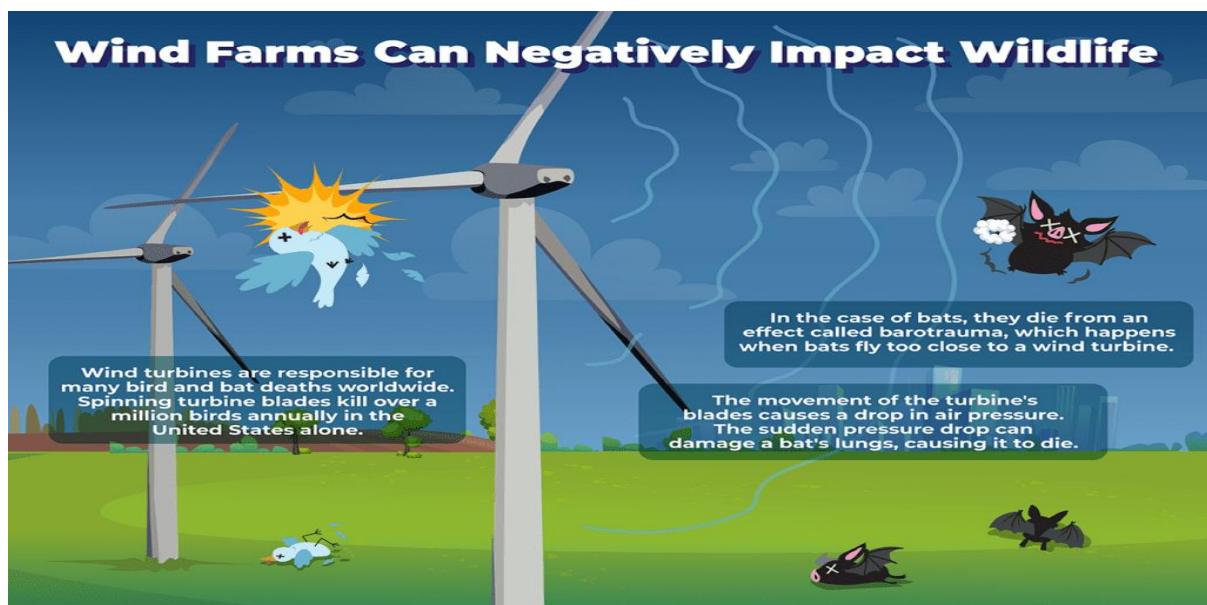
- A. Researchers investigated that the bird abundance is influenced by the number or power of turbines in a wind farm installation.
- B. This aids to determine how the bird population is affected by the scale of wind energy development.

4. Other Ecological Factors:

- A. Studies also recognised additional ecological factors or wind farm features that impact bird abundance. [4]

In a nutshell understanding the influence of wind turbines on birds is significant for stabilizing renewable energy targets with wildlife conservation. While wind energy mitigates climate change, decreasing its impact on bird populations remains a prime concern.

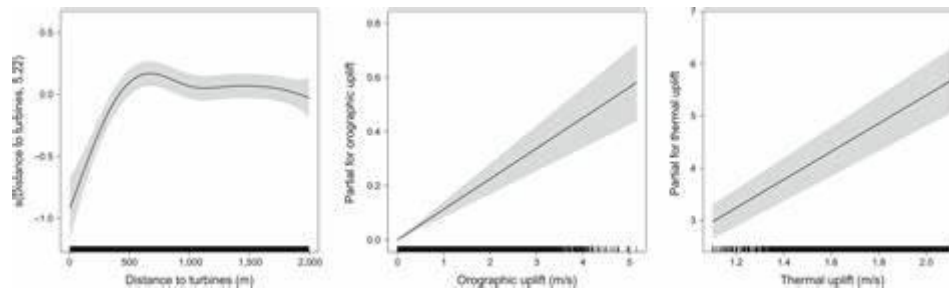
METHODOLOGY:



There are two principal ways of determining the climate influence of wind turbines. One is to correlate the original climate of a place after and before wind farms are installed there. This viewpoint is restricted by the accessibility of observations – new wind farms show relatively modern phenomenon so much data is not there to examine the climate after wind farms are designed with the years before. Applying this observational approach, researchers have discovered that the climate surrounding a large wind farm in a particular place was influenced by the existence of the turbines. The ground temperatures need to be measured by satellites; and they observed a warming of 0.5°C in the place directly under the farm at night. [7] This broiling effect was small and local – the warming during night did not enlarge far ahead the farm's nearest neighbourhood. But many people do not live right among the wind turbines. To look at miniature impacts at a great distance from the wind farms, another point of view is to affect the earth's climate with computer models. Applying climate history data, predictions are built that corresponds to thermodynamics, chemistry, motion and radiation in the sea, land and atmosphere. Simulations are run for years without and with wind farms installed. This method has the merit of having the ability to identify very small climate change signals, since the model can be run for a long duration, and natural variability usually decreases as it is averaged over more and more years. Also, every factor beside wind that can give rise to climate changes can be held persistent. [6] The demerit is that the climate behaviour model may not resemble exactly to cases that happens in real world. When this method was first executed the climate in their model was correlated with and without extraordinary large wind farms (huge enough to produce about double the global total current electrical demand), it was observed that in addition to climate effects in the prompt vicinity of the wind farms, there were alterations in climate all around the world. There were places of cooling and warming of about 0.5°C and decrease and increase in precipitation by a few percent. An analysis of the experiment showed that these alterations were usually a result of wind direction changes caused by the wind turbines. The model winds tended to shy away from the wind farms a little, so that downstream from the farms there would be regions of extra wind from the south, that would tend to be warmer, and regions of extra wind from the north, that would tend to be colder. [8]

1. Displacement Studies:

- A. Bird displacements are caused by wind turbines, where respective birds move permanently to other regions due to the existence of turbines.
- B. An organised analysis of 71 peer-reviewed studies on displacement observed the following drift:
 - 1. All most every observation was guided in Europe and North America, particularly in agricultural regions.
 - 2. Studies reported to displacement effects comprised approximate 40.6%.
 - 3. An increased abundance of birds around wind farm created about 7.7% detected attraction.
 - 4. Vigorous and uniform experimental designs are advised for upcoming research, along with in depth reporting of results, whether effects are absent, positive, or negative. [9]



2. Mitigation Strategies:

A. Wind turbine placement and design can help reduce negative effects on birds:

1. Shorter blades and taller turbines can decrease collision risk.
2. Positioning away from bird habitats is critical.

3. Monitoring and Assessment:

B. Researchers applied multiple procedures to test wind turbine impacts on birds:

1. Behaviour and population measurements observations near turbines.
2. Preconstruction surveillance to check wildlife activity levels.
3. Post-construction surveys to detect patterns and influence. [9]

Two methods are frequently used to determine a wind park's 'acceptable mortality limits'. In the excess mortality method (EMM) a minor increase in mortality above natural mortality is considered acceptable, as the impact is considered negligible (for example a 1% to 5% increase in mortality). Potential Biological Removal (PBR) is also examined; This term used to calculate the proportion of a population that can be removed without significant consequences for that population. Both methods are frequently used as a framework for issuing permits, in many parts of Europe. [10]

COMPUTATIONAL DATA:

1. Computational Fluid Dynamics (CFD) Modelling:

- Researchers applied CFD simulations to investigate the downstream wake effects of wind turbines. For example, in Chios Island in the North Aegean Sea, a study was conducted to explore wind farms and their wake effects using ANSYS Fluent. The analysis considered pressure, distribution, temperature and turbulence.

The determining equations of the flow field constitutes:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) = 0 \quad (1)$$

$$\frac{\partial}{\partial t} (\rho \vec{u}) + \nabla \cdot (\rho \vec{u} \vec{u}) = -\nabla p + \nabla \cdot (\bar{\tau}) + \rho \vec{g} + S_M \quad (2)$$

In Equations (1) and (2), ρ is the air density, $\vec{u}$ is the fluid velocity vector, $\vec{g}$ is the gravitational acceleration, p the pressure, $\bar{\tau}$ the shear stress tensor, and S_M a momentum source term. The stress tensor, is related to the strain rate as follows:

$$\bar{\tau} = \mu \left[\nabla \vec{u} + \nabla \vec{u}^T - \frac{2}{3} \cdot \nabla \cdot \vec{u} I \right] = 0 \quad (3)$$

where μ is the dynamic viscosity of the fluid, $\nabla \vec{u}$ denotes the velocity gradient, with T denoting the transpose of the matrix, and I is the unit tensor.

The modelling of turbulence at the Reynolds-Averaged Navier–Stokes (RANS) point of view expect the modelling and correlation of the Reynolds stresses to the mean velocity gradients. The Bossiness hypothesis is applied in the k – ε model representing the following relation. [11]

$$-\rho \overline{u'_i u'_j} = \mu_t \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \left(\rho k + \mu_t \frac{\partial u_k}{\partial x_k} \right) \delta_{ij}$$

where δ_{ij} is the Kronecker's delta, k is the turbulence kinetic energy, and μ_t the turbulent viscosity.

The k – ε turbulence closure scheme was appropriately parameterized for atmospheric flows. It assumes μ_t is an isotropic scalar quantity and it is computed as a function of the model transport equations for the turbulence kinetic energy, k and its dissipation rate, ε . The two additional transport equations are given below. [11]

$$\frac{\partial (\rho k)}{\partial t} + \frac{\partial (\rho k u_i)}{\partial x_i} = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + G_k + G_b - \rho \varepsilon - Y_m + S_k \quad (5)$$

$$\frac{\partial (\rho \varepsilon)}{\partial t} + \frac{\partial (\rho \varepsilon u_i)}{\partial x_i} = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon} \right) \frac{\partial \varepsilon}{\partial x_j} \right] + C_{1\varepsilon} \frac{\varepsilon}{k} (G_k + C_{3\varepsilon} G_b) - C_{2\varepsilon} \rho \frac{\varepsilon^2}{k} + S_\varepsilon \quad (6)$$

In Equations (5) and (6), G_k represents the generation of turbulence kinetic energy k due to the mean velocity gradients, G_b is the generation rate of k due to buoyancy, Y_m is the contribution of the fluctuating dilatation in compressible turbulence to the overall dissipation rate, and S_k and S_ε are source terms.

The turbulent (or eddy) viscosity, μ_t is computed by:

$$\mu_t = \rho C_\mu \frac{k^2}{\varepsilon} \quad (7)$$

where C_μ is a constant. The constants of the standard k – ε model are modified according to the analysis presented in detail in Prospathopoulos et al. [16] as follows:

$$C_\mu = 0.033, \quad C_{1\varepsilon} = 1.12, \quad C_{2\varepsilon} = 1.83, \quad \sigma_k = 1.0, \quad \sigma_\varepsilon = 1.3 \quad (8)$$

Modelling of atmospheric boundary:

$$G_b = -\mu_t \left(\frac{\partial u}{\partial z} \right)^2 \cdot \frac{Ri}{f_m}$$

$$Ri = \zeta \frac{0.74 + 4.7\zeta}{(1 + 4.7\zeta)^2}$$

$$f_m = 1 + 5\zeta$$

$$\zeta = \frac{z}{L}$$

where μ_t is the turbulent viscosity, Ri is the Richardson number, and L the Monin-Obukhov length.

$$u_0(z) = \frac{u_*}{\kappa} \left[\ln \left(\frac{z}{z_0} \right) + 5 \frac{z}{L} \right]$$

$$k_0(z) = \frac{u_*^2}{\sqrt{C_\mu}} \cdot \left(\frac{f_\varepsilon}{f_m} \right)^{0.5} \quad (14)$$

$$\varepsilon_0(z) = \frac{u_*^3}{\kappa \cdot z} \cdot f_\varepsilon \left(\frac{f_\varepsilon}{f_m} \right)^{0.5} \quad (15)$$

where $f_\varepsilon = 1 + 4 \frac{z}{L}$, u_* is the friction velocity, z is the vertical height, z_0 the roughness height, and $\kappa = 0.41$, the von Karman's constant.

2. High-Resolution Simulations:

- High-resolution simulations using models like the Weather Research and Forecasting (WRF) model help quantify downstream effects on near-surface climate. These simulations consider differences arising from various parameterizations. Wind turbines create wakes of slower, more choppy air that eventually spread and recover their momentum, impacting local climates.

3. Integrated Microclimate-Energy Demand Simulation:

- An integrated approach combines atmospheric, soil, vegetation, and building models to assess microclimate impacts. Understanding how wind turbines affect local conditions is crucial for sustainable energy planning. [12]

The quantification of the impacts of wind-power plants on birds at the order level is based on the generic concept of a species-area relationship (SAR), which is widely used in life cycle assessment (LCA). SARs predict the effect that area loss has on species richness. Species richness loss in LCIA is normally quantified in units of potentially disappeared fraction of species (PDF), which is a measure of the potential loss of species richness due to a given stressor. Different types of SARs are in use in LCA, such as the classical SAR, matrix-calibrated SAR or countryside SAR. The latter two account for habitat heterogeneity and the fact that species may survive in human-modified landscapes as well. We use the classical SAR for our purposes here, and instead introduce a probability of presence including a habitat-specific relative diversity. [12]

$$\frac{S_{new}}{S_{org}} = \left[\frac{A_{new}}{A_{org}} \right]^z \Leftrightarrow S_{lost} = S_{org} - S_{org} \cdot \left[\frac{A_{new}}{A_{org}} \right]^z \quad (1)$$

$$PDF = \frac{S_{lost}}{S_{org}} = \frac{S_{org} \cdot \left(1 - \left[\frac{A_{org} - A_{lost}}{A_{org}} \right]^z \right)}{S_{org}} \quad (2)$$

Quantitative assessment methods employed outside the LCIA framework include:

- A. Collision Risk Models: Predicting the risk of bird-turbine collisions.
- B. Species Distribution Models: Estimating species occurrences across regions.
- C. Individual-Based Models: Assessing individual bird behavior.
- D. Population Modeling Approaches: Scaling vulnerability to mortality, disturbance, and habitat alterations. [13]

1. Cumulative Impact Assessment:

- A. Combining various modeling techniques is promising for assessing cumulative impacts of wind farms on birds. Researchers consider both local and regional consequences.
- B. Addressing complex problems related to long-term impact assessments involves integrating these models. [13]

2. Specific Computational Models:

A. Stochastic Soaring Raptor Simulator:

- 1. Uses tracking information to predict long-distance flight paths of golden eagles. It can also estimate bird activity and habitat conditions in specific locations.

B. Other Computational Models:

- 1. Researchers have developed models to simulate thunderstorm downbursts for wind turbine load calculations and evaluate blade erosion induced by raindrop impact. [13]

1. **RESULTS:** Wind turbines create some near-surface warming by enlarging mixing of the lower atmosphere, almost identical to the influence of trees and high rise buildings. However, this impact is very small and mostly happens at night when temperatures increase above the ground. Researchers have studied that following the installation of a wind farm, there was a reduction in annual wind speed and an elevation in air and ground temperatures. Again, evapotranspiration showed a significant increasing drift. [14]

- a. Wind energy evolution can change microclimates via alterations in surface-atmosphere interchange of momentum, mass, energy, and trace gases. Studies have proved that:
 - i. Total bird species corresponds to 63%, 72% of bats, and 67% of terrestrial mammals' and displays ignorance behavior, driving clear of wind turbine regions.
 - ii. Ordinary cranes, reindeers, and owls, were mostly affected which exhibited consistent displacement, regularly withdrawing as far as 5 km from the turbines.
- b. The impact of rotor swept area and turbine height on bird and bat collision fatalities remains undetermined.
- c. There are adverse reports on whether bird and bat collisions increase with tower height or rotor swept area on a per megawatt basis. Habitat Loss for Soaring Birds:
- d. A novel study shows that the effect of wind power production on wildlife may be higher than previously acknowledged.
- e. A radius of about 650-700 meters around each and every wind turbine corresponds to flight space lost for soaring birds. [14]

2. Observational Studies:

- a. Scientist immediately measure and observe alterations in microclimate parameters near wind turbines.
- b. Curtailment Periods: During wind farm curtailment (when turbines are intentionally shut down), sudden changes heat and in turbulent fluxes of momentum can be measured.
- c. Temperature Changes: Wind turbines elevate mixing between near-surface and higher-atmosphere air, causing a change in air and land surface temperatures. [14]

3. Quantifying Wake Effects:

- a. Scientists' tests wake effects length using Monin–Obukhov similarity theory and actuator disk models.
- b. They evaluate how wind farms alter local-scale meteorology and microclimate.

4. Long-Term Monitoring:

- a. Monitoring continuously near wind turbines provides data on humidity, temperature, wind speed, and other microclimate parameters.
- b. Researchers compare data from areas without and with wind turbines to identify alterations.

5. Integrated Approaches:

- a. Integrating field measurements, modelling and remote sensing helps to understand the physical link between turbines and microclimate effects.
- b. Researchers studies alterations in turbulent fluxes, heat, and momentum.

Research associates used surplus mortality technique and potential biological removal techniques to check how, according to the rules, ‘acceptable mortality’ effects black-tailed godwit, starling, marsh harrier, stork, common spoonbill, common tern and sea eagle populations. The observations denotes that population size can be highly sensitive to minor increases in mortality (EMM). Instead of a negligible effect, research associates found that a 1% surplus mortality caused a 2% to 24% reduction in different bird populations after 10 years. A 5% elevation of the existing mortality resulted in a 9% to 77% decrease in the different populations after 10 years, depending on the species. This is particularly the case for starlings. Species with a short life-expectancy such as the starling already have high mortality figures, making them extremely sensitive to a percentage increase in excess mortality. [13]

When the PBR method is used to determine acceptable mortality limits, the change in numbers of birds appeared to be species-dependent and was largely determined by the recovery factor: a measure of population vulnerability. When this is 1, a value typically used for robust populations that can better withstand excess mortality, this excess mortality resulted in a 50% to 55% decrease in bird numbers. With a recovery factor of 0.1, an extremely conservative value used for endangered species, the decrease in bird numbers was approximately 5%. [13]

CONCLUSION:

Ultimately, it is now crystal clear that wind turbines have influences on both the birds and the microclimate. Some of the studies effects are mentioned in a encapsulated way:

1. Microclimate:
 - A. Due to their operation wind turbines introduce turbulences, affecting local mixing processes.
 - B. However, experiments recommended that this turbulence does not remarkably disturb temperature and pressure distributions downstream of wind farms.
 - C. Cooling happens in the close wake of turbines, while warming happens in the distant wake.
 - D. Overall, the effect on the microclimate seems to be nominal. [15]
2. Birds:
 - A. Wind turbines have an impact on bird distribution and abundance.
 - B. Vulnerability ranges among bird taxa, with few species more responsive to turbine effects.
 - C. Factors like power, turbine density, and ecological context play an important part.
 - D. An orientation between renewable energy aims with bird conservation remains critical. [15]

As a result, understanding these effects helps inform sustainable energy practices while minimizing disruption to local ecosystems and wildlife.

REFERENCE:

- 1) Sherman, P., Chen, X. & McElroy, M. Offshore wind: An opportunity for cost-competitive decarbonization of China's energy economy. *Sci. Adv.* **6**, 1–9 (2020).
- 2) Kaplan, Y. A. Overview of wind energy in the world and assessment of current wind energy policies in Turkey. *Renew. Sustain. Energy Rev.* <https://doi.org/10.1016/j.rser.2014.11.027> (2015).
- 3) Goodale, M. W. & Milman, A. Cumulative adverse effects of offshore wind energy development on wildlife. *J. Environ. Plan. Manag.* **59**, 1–21 (2016).
- 4) Drewitt, A. L. & Langston, R. H. Collision effects of wind-power generators and other obstacles on birds. *Ann. N. Y. Acad. Sci.* **1134**, 233–266 (2008).
- 5) Stergiannis, N.; van Beeck, J.; Runacres, M.C. Full HAWT rotor CFD simulations using different RANS turbulence models compared with actuator disk and experimental measurements. *Wind Energ. Sci. Discuss.* **2017**. preprint.
- 6) Cervarich, M.C.; Roy, S.B.; Zhou, L. Spatiotemporal Structure of Wind Farm-atmospheric Boundary Layer Interactions. *Energy Procedia* **2013**, *40*, 530–536.
- 7) Rados, K.; Mosfilis, S.; Stergiannis, N.; Tournlidakis, A.; Caralis, G.; Zervos, A. CFD modeling approaches of wind turbine single and multiple wakes. In Proceedings of the 2012 European Wind Energy Conference & Exhibition, Copenhagen, Denmark, 19 April 2012; pp. 1842–1849.
- 8) Everaert, J. Collision risk and micro-avoidance rates of birds with wind turbines in Flanders. *Bird Study* **61**, 220–230 (2014).
- 9) May, R. *et al.* Paint it black: Efficacy of increased wind turbine rotor blade visibility to reduce avian fatalities. *Ecol. Evol.* **10**, 8927–8935 (2020).
- 10) Tabassum-Abbasi, & Premalatha, M. & Abbasi, Tasneem & Abbasi, S.A., 2014. "Wind energy: Increasing deployment, rising environmental concerns," *Renewable and Sustainable Energy Reviews*, Elsevier, vol. 31(C), pages 270-288.
- 11) David De la Vega & James C. G. Matthews & Lars Norin & Itziar Angulo, 2013. "Mitigation Techniques to Reduce the Impact of Wind Turbines on Radar Services," *Energies*, MDPI, vol. 6(6), pages 1-15, June.
- 12) Walsh-Thomas, J.M., Cervone, G., Agouris, P., Manca, G. (2012). Further evidence of impacts of large-scale wind energy on land surface temperatures. *Renewable and Sustainable Energy Reviews*, *16*: 6432-6437.
- 13) Verberk, W.C.E.P., Leuven, R.S.E.W., van Duinen, G.A., Esselink, H. (2010). Loss of environmental heterogeneity and aquatic macroinvertebrate diversity following large-scale restoration management. *Basic and Applied Ecology*, *11*: 440-449.
- 14) Smith, J., Nayak, D.R. Smith, P. (2014). Wind farms on undegraded peatlands are unlikely to reduce future carbon emissions. *Energy Policy*, *66*: 585-591.
- 15) Santos, H., Rodrigues, L., Jones, G., Rebelo, H. (2013). Using species distribution modelling to predict bat fatality risk at wind energy. *Biological Conservation*, *157*: 178-186.